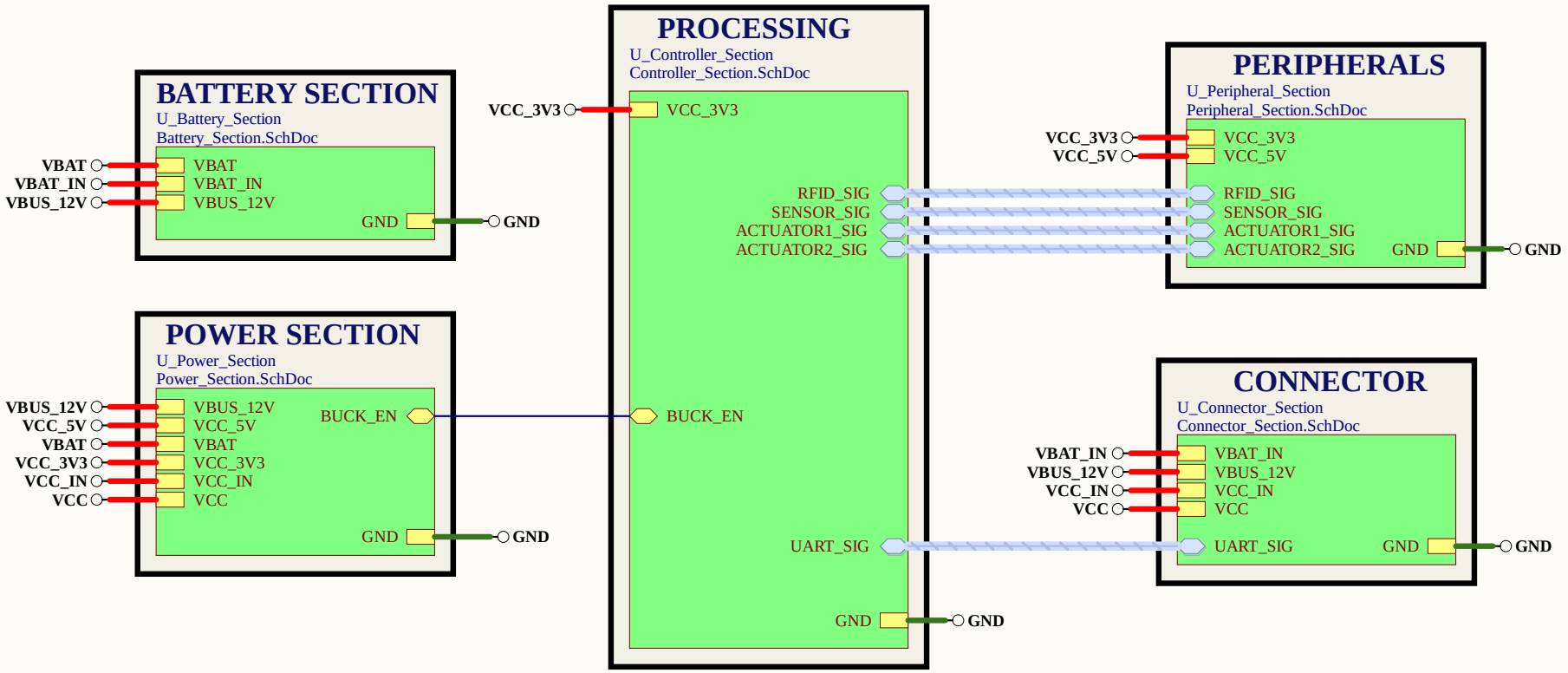
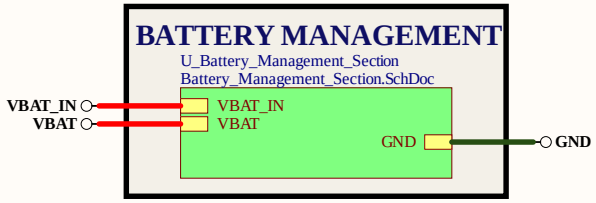
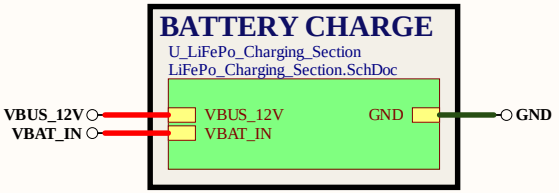
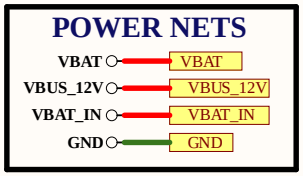
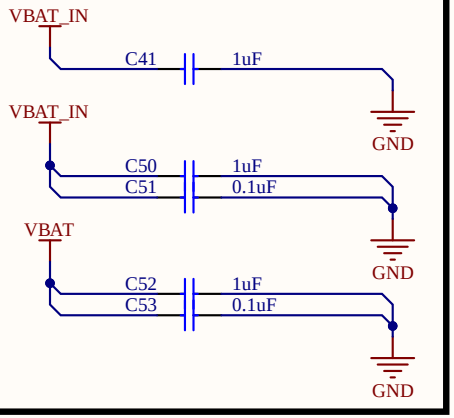

BATTERY SECTION
U_Battery_Section
Battery_Section.SchDoc
VBAT
VBAT_IN
VBUS_12V
GND
VBAT
VBAT_IN
VBUS_12V
GND
POWER SECTION
U_Power_Section
Power_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
BUCK_EN
GND
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
BUCK_EN
GND
PROCESSING
U_Controller_Section
Controller_Section.SchDoc
VCC_3V3
VCC_3V3
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
BUCK_EN
UART_SIG
GND
GND
PERIPHERALS
U_Peripheral_Section
Peripheral_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
GND
GND
CONNECTOR
U_Connector_Section
Connector_Section.SchDoc
VBAT_IN
VBUS_12V
VCC_IN
VCC
VBAT_IN
VBUS_12V
VCC_IN
VCC
UART_SIG
UART_SIG
GND
GND


BATTERY MANAGEMENT
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT_IN
VBAT
VBAT_IN
VBAT
GND
GND

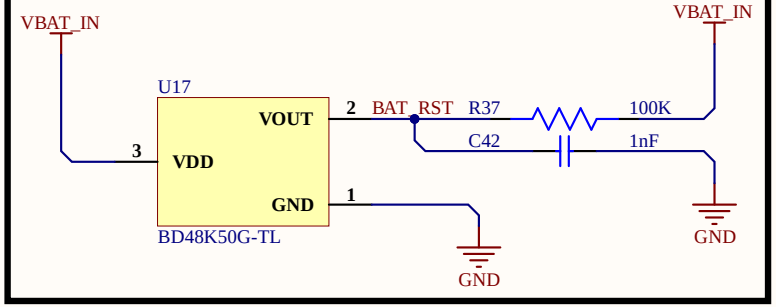

BATTERY CHARGE
U_LiFePo_Charging_Section
LiFePo_Charging_Section.SchDoc
VBUS_12V
VBAT_IN
VBUS_12V
VBAT_IN
GND
GND


POWER NETS
VBAT
VBUS_12V
VBAT_IN
GND
VBAT
VBUS_12V
VBAT_IN
GND

DECOUPLING CAPACITOR

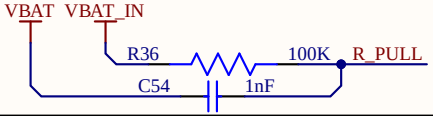


BATTERY SUPERVISORY CIRCUIT

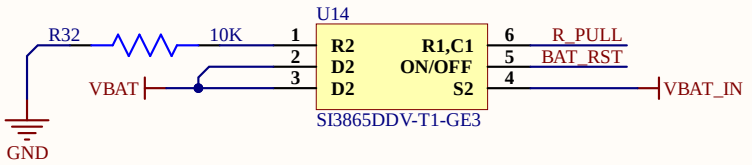


Detection voltage: 2.3V to 6.0V (Typ.)
Voltage threshold: 5V

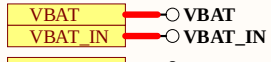
PULL UP RESISTOR



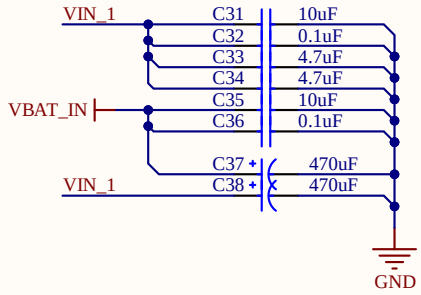
POWER SWITCH IC



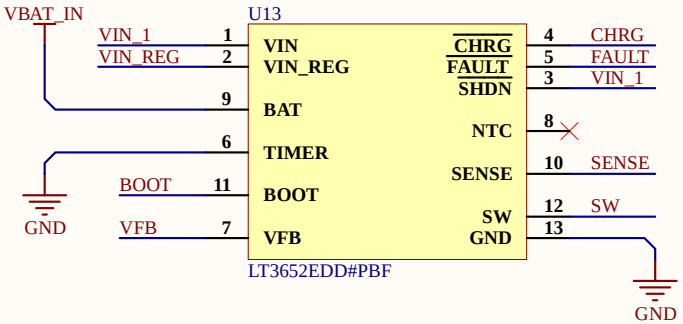
POWER NETS



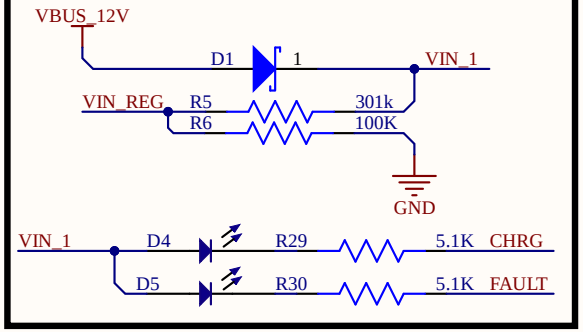
DECOUPLING CAPACITORS



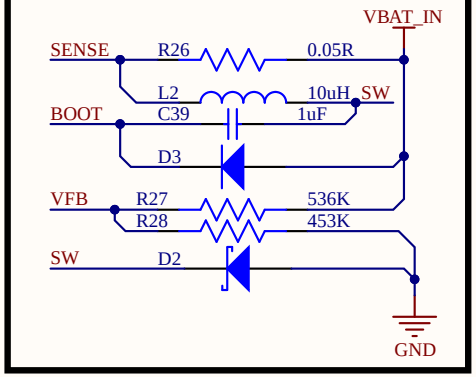
BATTERY CHARGING CIRCUIT



SUPPORTING CIRCUITRY

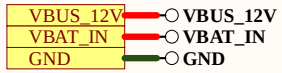


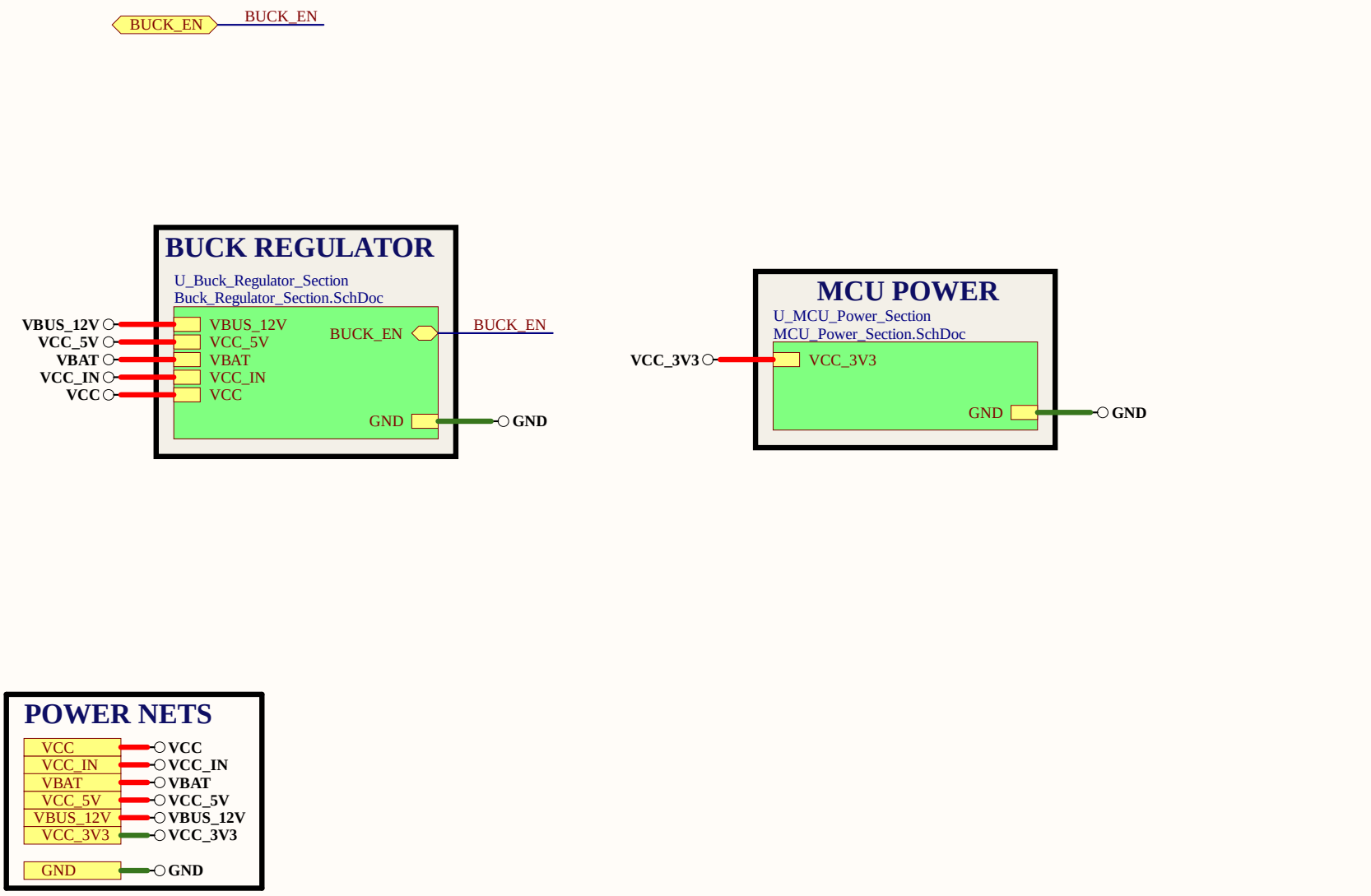
BATTERY INPUT CIRCUIT



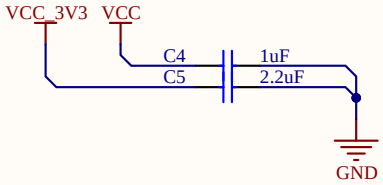
Battery Chemistry: 2 Cell LiFeOP4
Input Voltage: 12V USB
Battery Voltage: (2 × 3.6V) = 7.2V
Charge current: 2A

POWER NETS

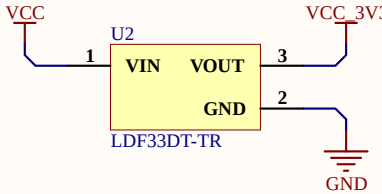



BUCK_EN
BUCK_EN
BUCK REGULATOR
U_Buck_Regulator_Section
Buck_Regulator_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
BUCK_EN
GND
BUCK_EN
GND
MCU POWER
U_MCU_Power_Section
MCU_Power_Section.SchDoc
VCC_3V3
VCC_3V3
GND
GND
POWER NETS
VCC
VCC_IN
VBAT
VCC_5V
VBUS_12V
VCC_3V3
GND
VCC
VCC_IN
VBAT
VCC_5V
VBUS_12V
VCC_3V3
GND

DECOUPLING CAPACITOR

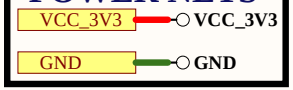


3V3 REGULATOR

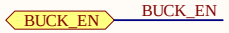


Input Voltage: 12V USB or 7.2V VBAT
Output Current: 1A
Output Voltage: Fixed 3.3V
Voltage Drop: 500 mV max. at 1 A load

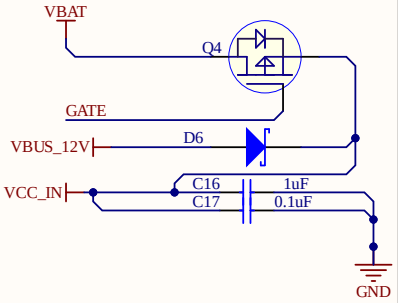
POWER NETS



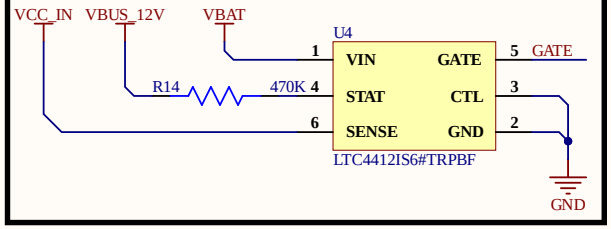
BUCK ENABLE



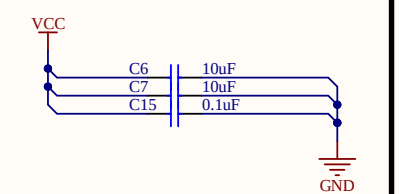
SUPPORTING CIRCUITRY



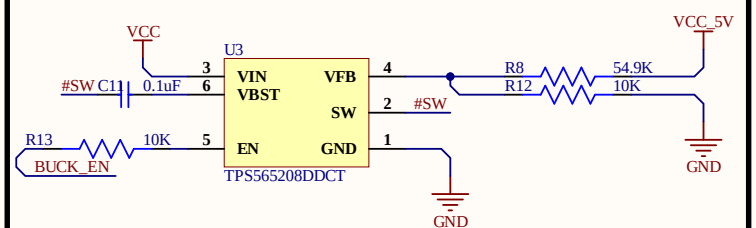
OR CONTROLLER SECTION



DECOUPLING CAPACITOR



BUCK SWITCHING REGULATOR CIRCUIT



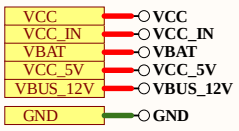
3.3uH inductor and output decoupling capacitor missing in the current design. (pg: 1,14.... datasheet)

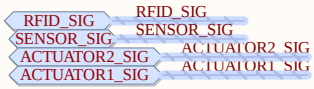
- Input Voltage: switches between 12V USB and VBAT (7.2V)
- Output Voltage: 5V
- Output Current: 5A Maximum.

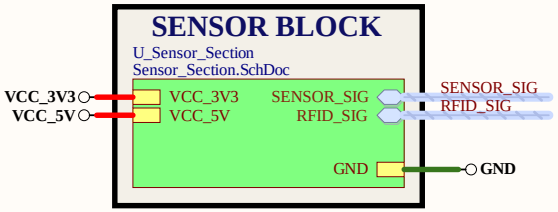
Output Voltage Calculation:

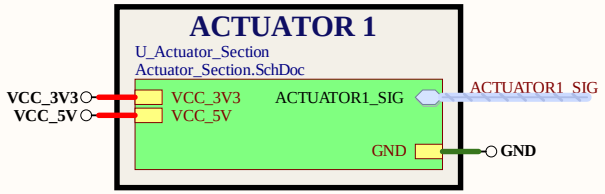
Formula: V =0.760 (1+R1/R2) (pg:13)
R1= 54.9K, R2=10K, L=3.3uH (pg:14)
VOUT= 4.93V
UVLO threshold: 3.6V (pg:5)

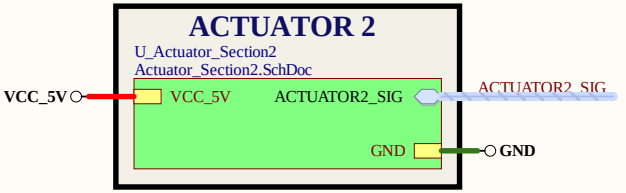
POWER NETS

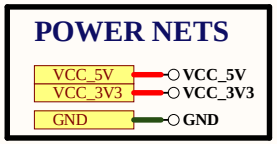


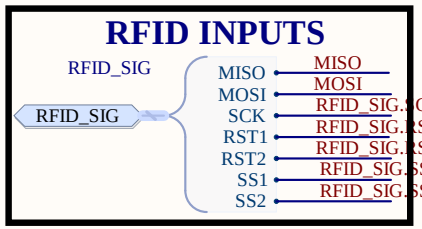

RFID_SIG
SENSOR_SIG
ACTUATOR2_SIG
ACTUATOR1_SIG
RFID_SIG
SENSOR_SIG
ACTUATOR2_SIG
ACTUATOR1_SIG

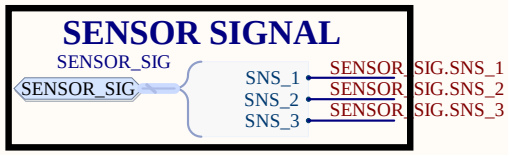

SENSOR BLOCK
U_Sensor_Section
Sensor_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
SENSOR_SIG
RFID_SIG
GND
SENSOR_SIG
RFID_SIG
GND


ACTUATOR 1
U_Actuator_Section
Actuator_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
ACTUATOR1_SIG
GND
ACTUATOR1_SIG
GND

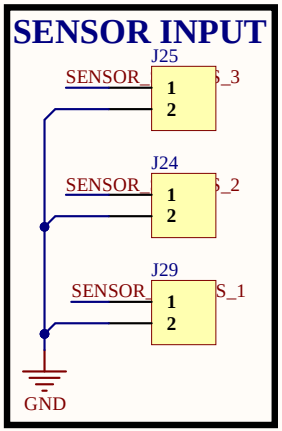

ACTUATOR 2
U_Actuator_Section2
Actuator_Section2.SchDoc
VCC_5V
VCC_5V
ACTUATOR2_SIG
GND
ACTUATOR2_SIG
GND

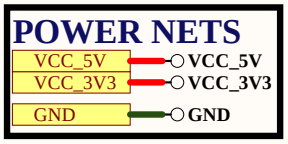

POWER NETS
VCC_5V
VCC_3V3
GND
VCC_5V
VCC_3V3
GND

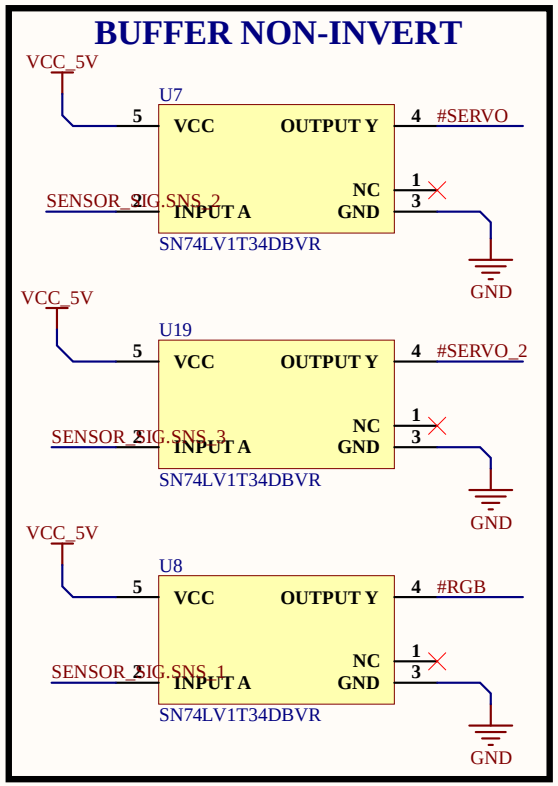


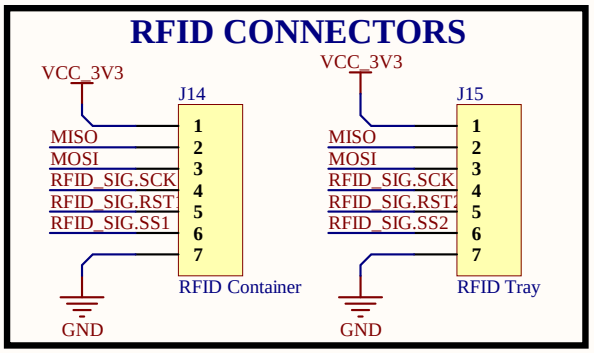




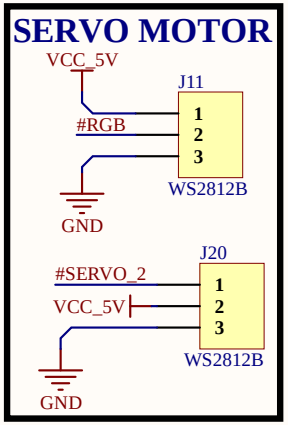


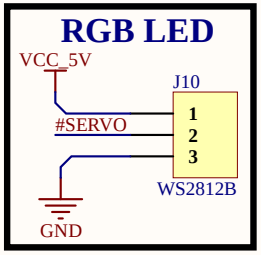


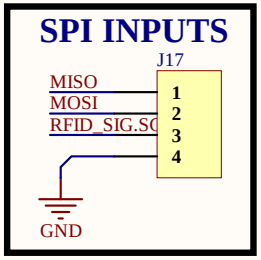


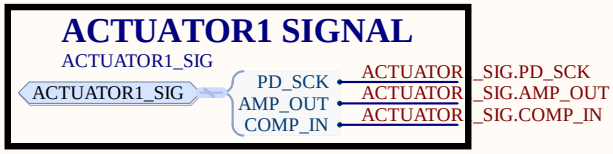


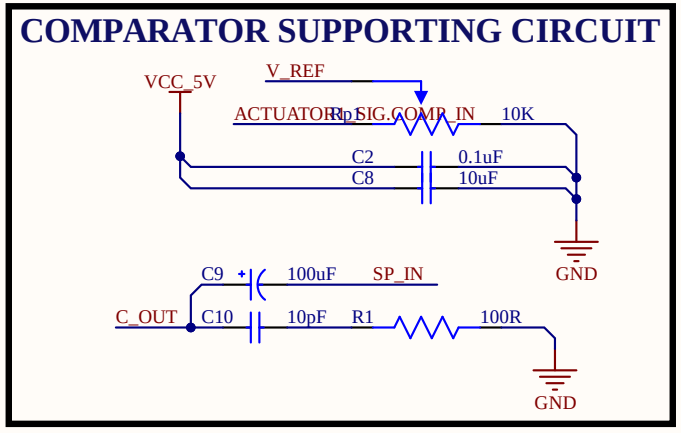
SNS signals are test signals for motor an d RGB LED.

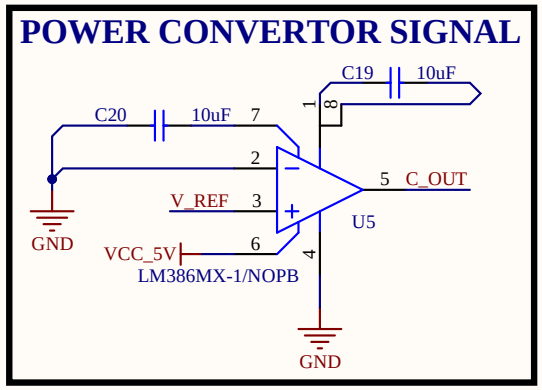


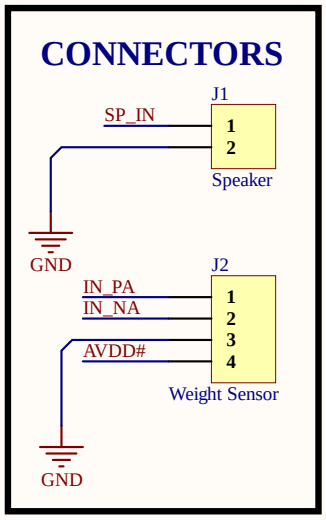


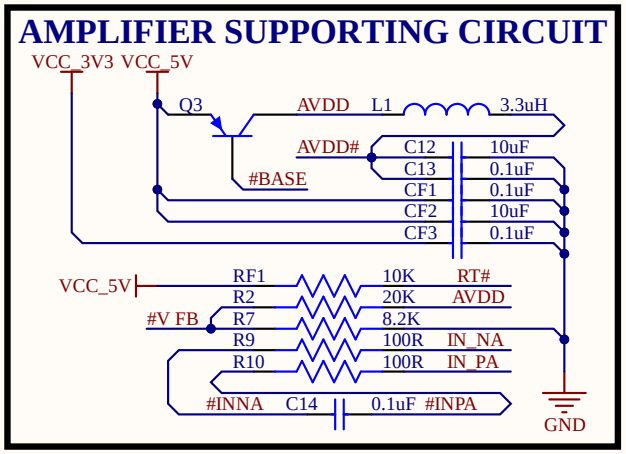


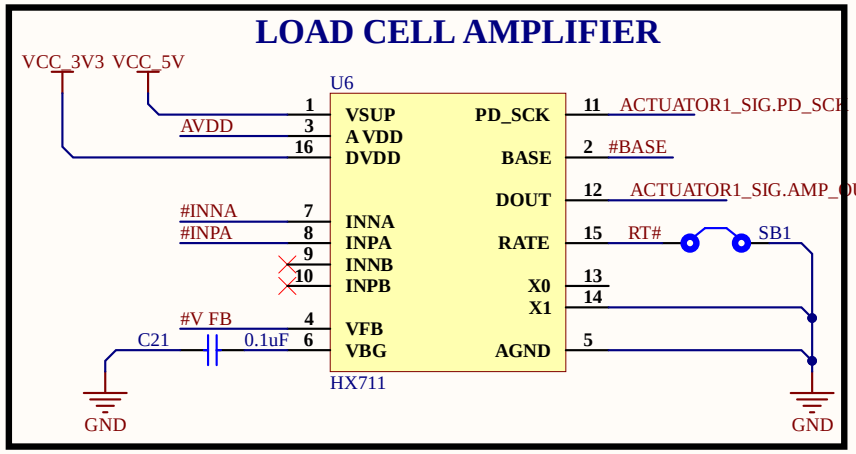

ACTUATOR1 SIGNAL
ACTUATOR1_SIG
ACTUATOR1_SIG
PD_SCK
AMP_OUT
COMP_IN
ACTUATOR_SIG.PD_SCK
ACTUATOR_SIG.AMP_OUT
ACTUATOR_SIG.COMP_IN

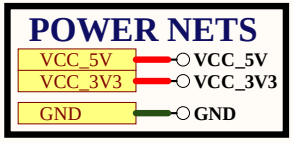

COMPARATOR SUPPORTING CIRCUIT
VCC_5V
V_REF
ACTUATOR_SIG.COMP_IN
R3
10K
C2
0.1uF
C8
10uF
GND
C9
100uF
SP_IN
C_OUT
C10
10pF
R1
100R
GND


POWER CONVERTOR SIGNAL
C19
10uF
C20
10uF
7
1
8
2
5
C_OUT
V_REF
3
U5
GND
VCC_5V
6
4
LM386MX-1/NOPB
GND

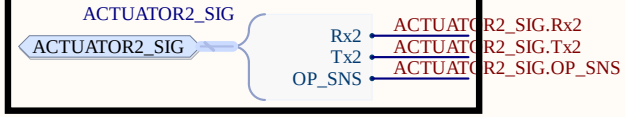

CONNECTORS
J1
SP_IN
1
2
Speaker
GND
J2
IN_PA
IN_NA
AVDD#
1
2
3
4
Weight Sensor
GND


AMPLIFIER SUPPORTING CIRCUIT
VCC_3V3
VCC_5V
Q3
AVDD
L1
3.3uH
AVDD#
C12
10uF
C13
0.1uF
#BASE
CF1
0.1uF
CF2
10uF
CF3
0.1uF
VCC_5V
RF1
10K
RT#
R2
20K
AVDD
#V FB
R7
8.2K
R9
100R
IN_NA
R10
100R
IN_PA
#INNA
C14
0.1uF
#INPA
GND

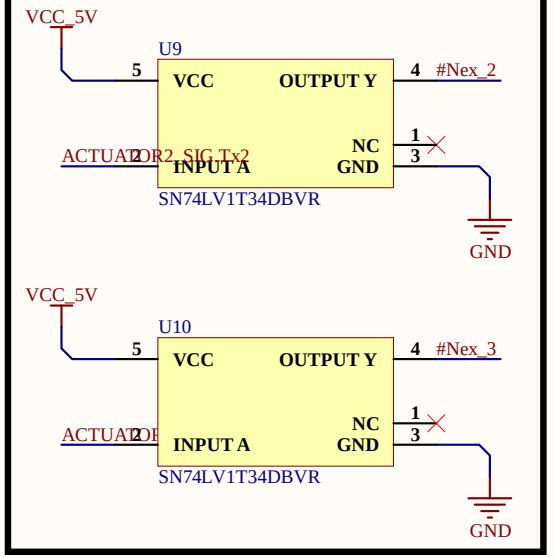

LOAD CELL AMPLIFIER
VCC_3V3
VCC_5V
U6
AVDD
1
3
16
VSUP
AVDD
DVDD
PD_SCK
BASE
DOUT
11
ACTUATOR1_SIG.PD_SCK
2
#BASE
12
ACTUATOR1_SIG.AMP_OUT
#INNA
#INPA
7
8
9
10
INNA
INPA
INNB
INPB
RATE
15
RT#
SB1
X0
X1
13
14
#V FB
4
C21
0.1uF
6
VFB
VBG
AGND
5
HX711
GND
GND


POWER NETS
VCC_5V
VCC_3V3
GND
VCC_5V
VCC_3V3
GND

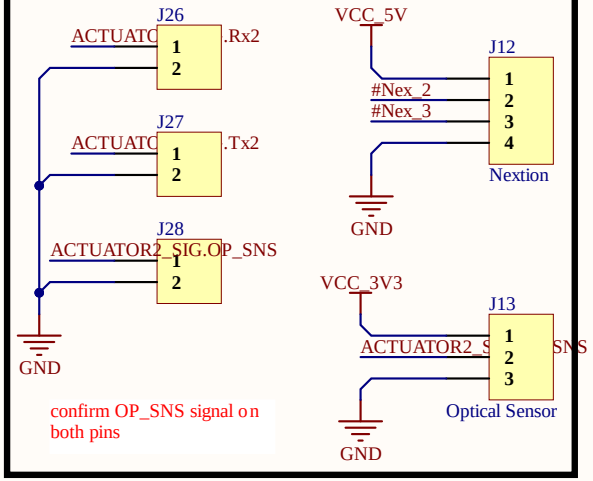
ACTUATOR2 SIGNAL



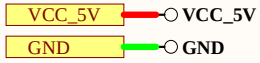
BUFFER CIRCUIT

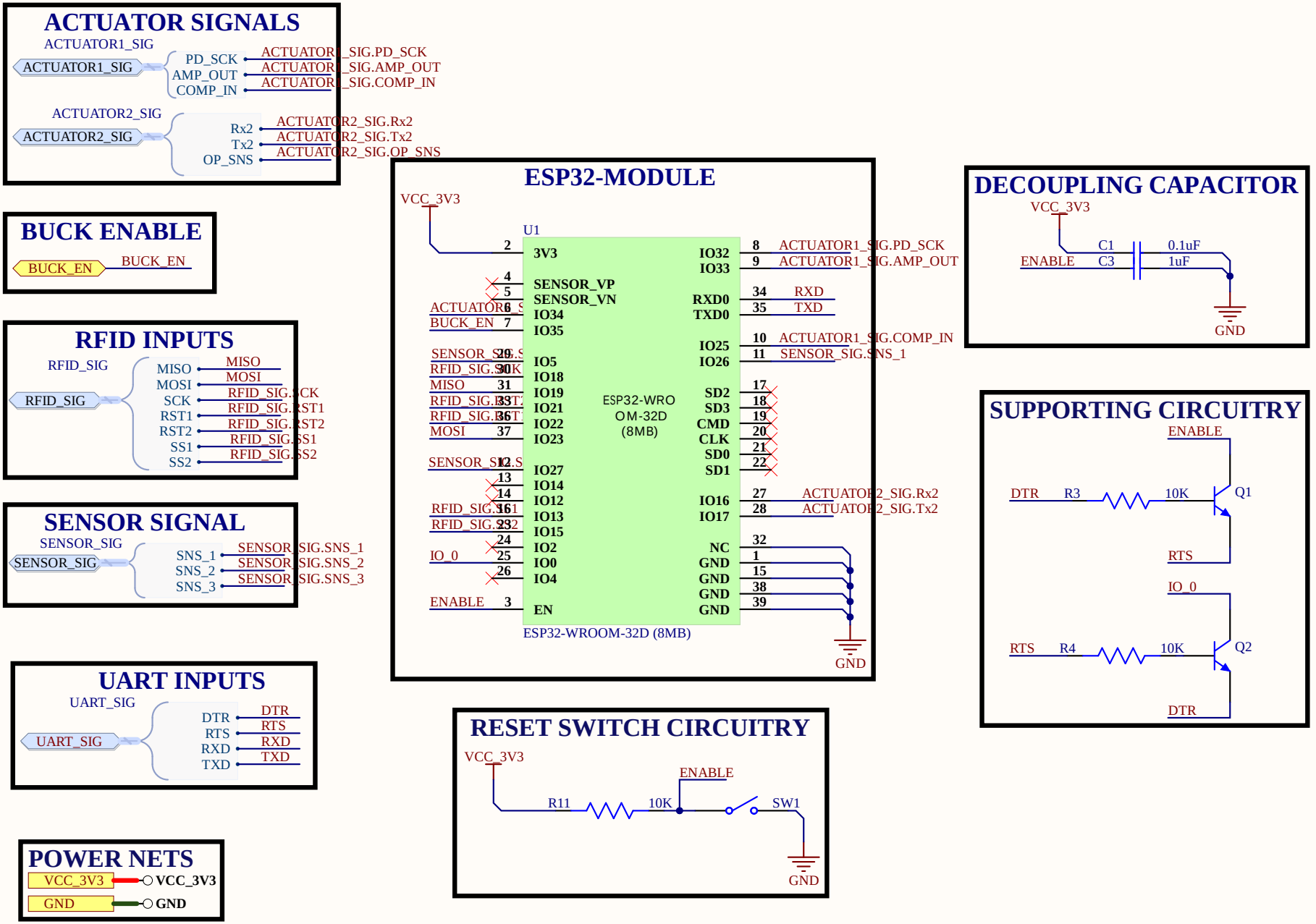


SENSOR INPUT CONNECTORS

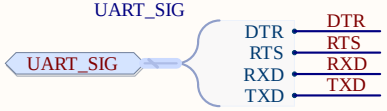


POWER NETS

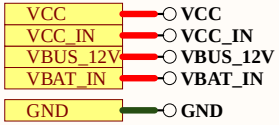



ACTUATOR SIGNALS
ACTUATOR1_SIG
ACTUATOR1_SIG
PD_SCK
AMP_OUT
COMP_IN
ACTUATOR1_SIG.PD_SCK
ACTUATOR1_SIG.AMP_OUT
ACTUATOR1_SIG.COMP_IN
ACTUATOR2_SIG
ACTUATOR2_SIG
Rx2
Tx2
OP_SNS
ACTUATOR2_SIG.Rx2
ACTUATOR2_SIG.Tx2
ACTUATOR2_SIG.OP_SNS
BUCK ENABLE
BUCK_EN
BUCK_EN
RFID INPUTS
RFID_SIG
RFID_SIG
MISO
MOSI
SCK
RST1
RST2
SS1
SS2
MISO
MOSI
RFID_SIG.SCK
RFID_SIG.RST1
RFID_SIG.RST2
RFID_SIG.SS1
RFID_SIG.SS2
SENSOR SIGNAL
SENSOR_SIG
SENSOR_SIG
SNS_1
SNS_2
SNS_3
SENSOR_SIG.SNS_1
SENSOR_SIG.SNS_2
SENSOR_SIG.SNS_3
UART INPUTS
UART_SIG
UART_SIG
DTR
RTS
RXD
TXD
DTR
RTS
RXD
TXD
POWER NETS
VCC_3V3
VCC_3V3
GND
GND
ESP32-MODULE
VCC_3V3
U1
2
3V3
4
5
SENSOR_VP
SENSOR_VN
6
IO34
BUCK_EN
7
IO35
20
IO5
30
IO18
MISO
31
IO19
33
IO21
36
IO22
MOSI
37
IO23
12
IO27
13
IO14
14
IO12
16
IO13
23
IO15
24
IO2
IO_0
25
IO0
26
IO4
ENABLE
3
EN
ESP32-WROOM-32D (8MB)
ESP32-WRO
OM-32D
(8MB)
IO32
IO33
RXD0
TXD0
IO25
IO26
SD2
SD3
CMD
CLK
SD0
SD1
IO16
IO17
NC
GND
GND
GND
GND
8
9
34
35
10
11
17
18
19
20
21
22
27
28
32
1
15
38
39
ACTUATOR1_SIG.PD_SCK
ACTUATOR1_SIG.AMP_OUT
RXD
TXD
ACTUATOR1_SIG.COMP_IN
SENSOR_SIG.SNS_1
ACTUATOR2_SIG.Rx2
ACTUATOR2_SIG.Tx2
GND
DECOUPLING CAPACITOR
VCC_3V3
ENABLE
C1
C3
0.1uF
1uF
GND
SUPPORTING CIRCUITRY
ENABLE
DTR
R3
10K
Q1
RTS
IO_0
RTS
R4
10K
Q2
DTR
RESET SWITCH CIRCUITRY
VCC_3V3
ENABLE
R11
10K
SW1
GND

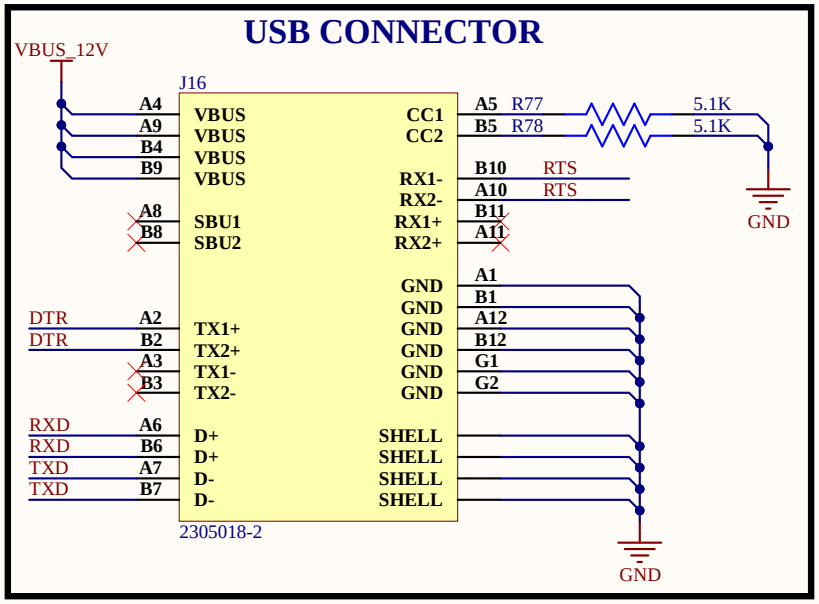
UART INPUTS



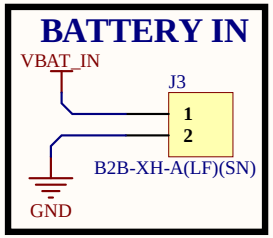
POWER NETS



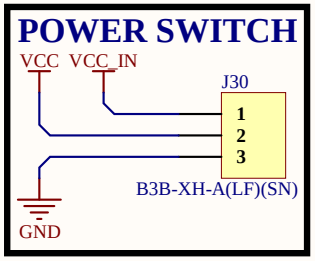
USB CONNECTOR



BATTERY IN



POWER SWITCH



VCC and VCC_IN are externally short circuited via a jumper on the board. VCC_IN is the main power input from source selector IC.